Machine Learning — Lecture 5

Model Evaluation, Validation, and Selection

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Lecture 5 at a Glance

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Model Evaluation, Validation, and Selection

Course Learning Outcome

CLO5: Evaluate machine learning models using appropriate metrics and justify decisions based on performance trade-offs and real-world context.

Lecture purpose

- Move from “model output” to “model quality”.
- Evaluate regression and classification models correctly.
- Understand validation, cross-validation, and test discipline.
- Interpret **class imbalance**, thresholds, and error costs.
- Select models based on both metrics and application context.

Why this lecture matters

A model is not useful only because it runs.

It is useful when we can answer:

- How well does it generalize?
- Which errors does it make?
- Which metric matters for the application?
- Which model should be selected?
- What is the cost of a wrong decision?

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💡 Main idea: evaluation is not only a calculation. It is a decision process that connects model behavior to the real-world problem.

Learning Objectives

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By the end of Lecture 5, you will be able to:

1. Distinguish training, validation, and test evaluation.
2. Explain why test data should not guide model selection.
3. Compute and interpret **regression metrics**: MAE, MSE, RMSE, and R^2.
4. Build and interpret a confusion matrix.
5. Compute accuracy, precision, recall, specificity, and F_1-score.
6. Explain why accuracy can be misleading under class imbalance.
7. Analyze threshold effects on false positives and false negatives.
8. Interpret ROC and precision-recall trade-offs conceptually.
9. Use cross-validation to support model selection.
10. Choose a model using performance metrics and real-world error costs.

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Lecture Roadmap

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From scores to responsible decisions

Learning phase	Parts	Guiding question
1. Establish evaluation discipline	**I**	Why is training performance not enough, and what makes an evaluation trustworthy?
2. Measure model performance	**II–IV**	Which regression and classification metrics reveal the errors that matter?
3. Make threshold and cost decisions	**V**	How do decision thresholds change outcomes and practical consequences?
4. Validate, select, and apply	**VI–VIII**	How do we compare models fairly and justify a final choice in context?

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💡 **Practice & consolidation:** compute metrics, inspect trade-offs, recognize evaluation mistakes, and defend a model choice.

Part I

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The Evaluation Problem

Building on Lectures 1–4, we now focus on **which evidence** supports a trustworthy evaluation and a defensible model choice.

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Why Training Accuracy Is Not Enough

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Memorization is not learning

A model can achieve **excellent training performance** by learning genuine patterns, but also by exploiting:

- noise and accidental correlations;
- duplicated examples;
- leaked information;
- sample-specific quirks.

The key distinction

Not sufficient
Low training error

≠

The real goal
Low error on future unseen data

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💡 **Takeaway.** Evaluation on unseen data estimates **generalization**, rather than merely confirming that the model remembers its training examples.

Development vs. Final Evaluation

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Different data have different roles

Dataset split	Main role	Should it guide model choice?
Training	fit parameters	yes, through learning
Validation	select model/hyperparameters	yes
Test	final unbiased estimate	no

Correct discipline

1. Train models on the training set.
2. Compare alternatives using the validation set.
3. Select the final model.
4. Evaluate once on the test set.

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If the test set influences repeated model choices, it becomes a validation set and no longer gives a clean final estimate.

Data Leakage in Evaluation

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When evaluation becomes too optimistic

Data leakage occurs when information from validation/test data enters training or model selection.

Common forms

Leakage source	Why it is problematic
Scaling before splitting	test distribution influences preprocessing
Feature selection using all data	test information affects chosen features
Repeated tuning on the test set	test set becomes part of development
Future variables used as features	prediction uses information unavailable at decision time
Duplicate examples across splits	model may see the same case during training

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> A model affected by leakage may look excellent in evaluation and fail in real deployment.

What Should Evaluation Answer?

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Not one question, but several

A useful evaluation should answer:

1. How large are typical errors?
2. Are there rare but severe errors?
3. Which classes or groups are hardest?
4. Does the model generalize beyond training data?
5. Is performance stable across folds or splits?
6. Which error type is most costly?
7. Is the model good enough for the real decision?
8. Is it better than a baseline?

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> A 95% accurate model can still be useless if it misses the rare event that matters most.

Baselines

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Every model should beat a simple reference

A **baseline** is a simple reference model.

Examples

Task	Possible baseline
Regression	always predict the training mean
Binary classification	always predict the majority class
Multiclass classification	predict the most frequent class
Time-dependent data	predict the last observed value
Ranking	random or popularity-based ranking

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> 💡 A complex model that barely beats a simple baseline may not justify its complexity.

Part II

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Regression Metrics

Regression evaluation measures how far numerical predictions are from actual values.

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Regression Error Notation

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Actual, prediction, residual

For each example i, compare the **actual value** with the model's **prediction**.

Quantity	Notation	Meaning
Actual value	y_i	Observed target
Predicted value	$\hat{y}_i$	Model prediction
Residual	$e_i = y_i - \hat{y}_i$	Signed difference
Absolute error	$\lvert e_i \rvert$	Magnitude of the error
Squared error	e_i^2	Penalizes larger errors more strongly

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Interpreting the residual

- $e_i > 0$: The model underpredicts.
- $e_i < 0$: The model overpredicts.
- $e_i = 0$: The prediction is exact.

💡 **Key idea.** Regression metrics summarize errors differently: **MAE** averages absolute errors, while **MSE** averages squared errors.

Mean Absolute Error

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Average size of the error

Mean Absolute Error is:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Interpretation

MAE is the average absolute **prediction error** in the same unit as the target.

Strengths

- easy to explain;
- less sensitive to outliers than squared-error metrics;
- useful when error cost grows approximately linearly.

Limitation

- does not strongly emphasize rare large mistakes.

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Mean Squared Error and RMSE

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Penalizing large errors

Mean Squared Error:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Root Mean Squared Error:

$$RMSE = \sqrt{MSE}$$

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Interpretation

- MSE gives large errors much more influence.
- RMSE returns to the original unit of the target.

> 💡 If large mistakes are especially costly, **RMSE** may be more informative than MAE.

Worked Example: Regression Metrics

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Four predictions

Example	Actual y	Prediction $\hat{y}$	Error e	$\lvert e \rvert$	e^2
1	100	90	10	10	100
2	120	130	-10	10	100
3	80	85	-5	5	25
4	200	160	40	40	1600

Then:

$$MAE = \frac{10 + 10 + 5 + 40}{4} = 16.25.$$

$$MSE = \frac{100 + 100 + 25 + 1600}{4} = 456.25.$$

$$RMSE = \sqrt{456.25} \approx 21.36.$$

The large error of 40 strongly increases MSE and **RMSE**.

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Coefficient of Determination

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R^2

The coefficient of determination compares prediction error with a mean-based baseline:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

- **SSE (Sum of Squared Errors):** $\sum_i (y_i - \hat{y}_i)^2$ — squared prediction errors.
- **SST (Total Sum of Squares):** $\sum_i (y_i - \bar{y})^2$ — total variation around the actual mean $\bar{y}$.

Interpretation

Value	Meaning
1	perfect predictions on evaluated data
0	same squared error as predicting the evaluated sample mean
< 0	worse than that mean-based prediction
close to 1	much less squared error than the mean baseline

A high training R^2 can still indicate overfitting if validation/test performance is weak.

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Worked Example: Interpreting R^2

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Use the same four regression predictions

From the previous example, $y = (100, 120, 80, 200)$ and $\hat{y} = (90, 130, 85, 160)$.

Mean actual value: $\bar{y} = 125$

$$SSE = 100 + 100 + 25 + 1600 = 1825$$

$$SST = (100 - 125)^2 + (120 - 125)^2 + (80 - 125)^2 + (200 - 125)^2 = 8300$$

$$R^2 = 1 - \frac{1825}{8300} \approx 0.780$$

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> The predictions reduce squared error by about **78%** relative to a constant prediction using the evaluated sample mean. This does **not** mean that 78% of future individual predictions are correct.

Choosing a Regression Metric

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Match metric to context

Application	Metric emphasis	Reason
Rent prediction	MAE	easy to interpret in currency
Medical dosage	RMSE or safety-specific metric	large errors may be dangerous
Energy forecasting	MAE + RMSE	typical and extreme errors both matter
Scientific model fit	R^2 plus residual analysis	explains variation but not all error behavior
Business forecasting	MAE, RMSE, percentage error	cost interpretation matters

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💡 There is no universally best metric. The metric should match the consequence of **prediction error**.

Residual Analysis

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Metrics are not the whole story

Residuals ($e_i = y_i - \hat{y}_i$) reveal patterns that aggregate regression metrics may hide.

Systematic patterns

- **Underprediction:** Residuals are consistently positive.
- **Overprediction:** Residuals are consistently negative.
- **Nonlinear structure:** Residuals follow a curved pattern.

Error variability

- **Increasing spread:** Error variance changes across predictions.
- **Outliers:** A few examples have unusually large residuals.
- **Group differences:** Errors are concentrated in particular groups.

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Example

If a model consistently **underpredicts expensive apartments**, it may fail to capture pricing patterns in the high-end market, even when its overall MAE is low.

> 💡 **Key idea.** Regression metrics summarize **how much** error occurs; residual analysis reveals **where and how** the model makes errors.

Visualize the Regression Errors

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One large residual changes RMSE strongly



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💡 The fourth prediction has the largest **residual** ($+40$). Squaring its error contributes **1600 of the total 1825 squared-error units**.

Part III

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Classification Metrics

Classification evaluation begins by counting the types of correct and incorrect decisions.

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Confusion Matrix

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Four outcomes for binary classification

Assume class 1 is the positive class.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Meaning

Term	Interpretation
TP	positive case correctly detected
TN	negative case correctly rejected
FP	false alarm
FN	missed positive case

> In fraud detection, FN means fraud was missed; FP means a legitimate transaction was flagged.

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Visual Confusion Matrix

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Connect the four cells to the metrics

200 labeled examples · positive class = 1

Actual	Predicted: Positive	Predicted: Negative
positive	**TP = 40** correct detection	**FN = 20** missed positive
negative	**FP = 10** false alarm	**TN = 130** correct rejection

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💡 In this example, **recall = 40/60** asks how many actual positives were detected, while **precision = 40/50** asks how reliable positive alerts were.

Accuracy

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Overall fraction correct

Accuracy measures the proportion of all predictions that are correct:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Most useful when

- Classes are reasonably **balanced**.
- Different errors have **similar costs**.
- Overall correctness is the main concern.

Can be misleading when

- One class is **much more common**.
- False positives and false negatives have **different costs**.
- Minority-class errors are hidden.

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💡 **Key idea.** Accuracy answers **“How often is the model correct?”** but not **“Which mistakes does it make?”**

Precision

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Of predicted positives, how many are correct?

Precision measures the reliability of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

What does high precision mean?

When the model predicts **class 1**, it is usually correct.

When is precision important?

When **false positives** are costly, for example:

- Incorrectly flagging legitimate transactions as fraud.
- Triggering unnecessary medical alarms.
- Blocking legitimate users.
- Labeling normal behavior as suspicious.

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💡 **Key idea.** Precision emphasizes **avoiding false alarms**. It does not tell us how many actual positive cases were missed.

Recall

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Of actual positives, how many are found?

Recall, also called **sensitivity** or **true positive rate**, measures how many actual positive cases the model detects:

$$\text{Recall} = \frac{TP}{TP + FN}$$

What does high recall mean?

The model successfully identifies **most actual positive cases**.

When is recall important?

When **false negatives** are costly, for example:

- Missing a disease during screening.
- Failing to detect fraudulent transactions.
- Missing students who need academic support.
- Failing to identify a safety threat.

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💡 **Key idea.** Recall emphasizes **not missing positive cases**. High recall does not necessarily mean few false alarms.

Specificity

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Of actual negatives, how many are identified?

Specificity, also called **true negative rate**, measures the proportion of actual negative cases correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

What does high specificity mean?

The model correctly identifies **most actual negative cases**, producing relatively few false positives.

When is specificity important?

When incorrectly flagging negative cases causes:

- Unnecessary investigations or alerts.
- Additional cost or workload.
- Harmful interventions.

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💡 **Key idea.** Raising the decision threshold generally **increases or preserves specificity**, but may **reduce recall**. The right balance depends on the application.

F1-Score

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Balancing precision and recall

The F_1**-score** is the harmonic mean of precision and recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

When is F_1 useful?

- The **positive class** is important.
- The dataset is **imbalanced**.
- Both **FP and FN** matter.
- One summary of precision and recall is needed.

Important limitations

- **Ignores true negatives**.
- Weights precision and recall **equally**.
- May not reflect actual **error costs**.
- Undefined cases need a convention, commonly setting the score to zero.

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💡 **Key idea.** A high F_1 requires **both precision and recall** to be high; a low value of either reduces the score.

Worked Example: Confusion Matrix

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Calculate the main metrics

Given: $TP = 40, \quad FP = 10, \quad FN = 20, \quad TN = 130$

Metric	Calculation
Accuracy	$\dfrac{40 + 130}{40 + 10 + 20 + 130} = \dfrac{170}{200} = 0.850$
Precision	$\dfrac{40}{40 + 10} = 0.800$
Recall	$\dfrac{40}{40 + 20} \approx 0.667$
Specificity	$\dfrac{130}{130 + 10} \approx 0.929$
F_1 **score**	$2 \times \dfrac{0.80 \times 0.667}{0.80 + 0.667} \approx 0.727$

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💡 **Interpretation.** The model correctly identifies most negatives (**92.9% specificity**) but misses **one-third of actual positives** (**66.7% recall**).

Which Metric Matters?

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Match the metric to the error cost

Application	Main concern	Metric emphasis
Cancer screening	Missing a disease (FN)	**Recall / sensitivity**
Spam filtering	Losing legitimate email (FP)	**Precision**
Fraud detection	Missing fraudulent activity (FN)	**Recall + cost analysis**
Search results	Returning irrelevant results (FP)	**Precision**
Safety monitoring	Missing dangerous events (FN)	**Recall**
Manual review queue	Overloading reviewers (FP)	**Precision + workload**

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💡 **Key lesson.** There is no universally best metric. Choose metrics based on **which errors matter most**, the **class distribution**, and the **real-world consequences** of decisions.

Part IV

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Class Imbalance

When one class is much more common than another, **high accuracy can hide poor performance on the minority class**.

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Majority-Class Baseline

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High accuracy does not guarantee a useful model

Suppose a dataset contains **1,000 transactions**:

- **990 legitimate** transactions (99%).
- **10 fraudulent** transactions (1%).

A **majority-class baseline** predicts every transaction as legitimate.

What happens?

$$\text{Accuracy} = \frac{990}{1000} = \boxed{99\%}$$

$$\text{Fraud Recall} = \frac{0}{0 + 10} = \boxed{0\%}$$

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Despite its high accuracy, the model **fails to detect any fraud**.

> **Key lesson.** Always compare against a **majority-class baseline** and examine **minority-class performance**, not accuracy alone.

Imbalance Changes Interpretation

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High accuracy may not be impressive

Suppose **98% of cases are negative**. A model that always predicts **negative** achieves **98% accuracy**, but **0% recall** for the positive class.

What should we evaluate?

Classification performance

- Confusion matrix;
- Precision;
- Recall;
- F_1 score.

Errors and decisions

- Class-specific errors;
- Decision-threshold behavior;
- Costs of FP and FN;
- Precision–recall curve, when appropriate.

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Examples of imbalanced tasks

Fraud detection · Rare disease diagnosis · Equipment failure prediction · Intrusion detection · At-risk student identification

Key idea. When the positive class is rare but important, evaluate **precision, recall, and the costs of false positives and false negatives**. High accuracy alone may hide poor detection of the cases that matter.

Balanced Accuracy

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Giving both classes equal importance

Balanced accuracy averages the recall of the positive class (**sensitivity**) and the negative class (**specificity**):

$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2}$$

Unlike ordinary accuracy, it gives **equal weight to both classes**, making it especially useful for **imbalanced datasets**.

Worked example

Suppose **Recall = 0.60** and **Specificity = 0.95**.

$$\text{Balanced Accuracy} = \frac{0.60 + 0.95}{2} = \boxed{0.775 = 77.5\%}$$

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💡 **Key idea.** Balanced accuracy prevents a large majority class from dominating the evaluation, but it does not directly account for the **different costs of errors**.

Part V

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Thresholds and Trade-offs

Many classifiers produce **probabilities or scores**. The **decision threshold** determines which predictions are labeled positive and controls the trade-off between **false positives and false negatives**.

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Threshold Movement

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Lower or Higher Threshold?

For predicted probability $p = P(y = 1 \mid x)$, predict **class 1** when $p \geq t$.

Lower Threshold

- **More positive predictions**.
- **Recall** increases or stays unchanged.
- **False positives** may increase.
- **Precision** may decrease, but not necessarily.

Higher Threshold

- **Fewer positive predictions**.
- **Recall** decreases or stays unchanged.
- **False negatives** may increase.
- **Precision** may improve, but not necessarily.

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💡 **Key idea.** Lower thresholds favor **detecting positives**, while higher thresholds favor **fewer alerts**. Choose the threshold using **validation data and real-world error costs**.

Threshold Example

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Predicted fraud probabilities

Transaction	True class	$P(\text{fraud})$
A	Fraud	0.91
B	Legitimate	0.64
C	Fraud	0.48
D	Legitimate	0.22
E	Fraud	0.18
F	Legitimate	0.07

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At threshold $t = 0.5$

Predicted positives: $A,\ B$

At threshold $t = 0.2$

Predicted positives: $A,\ B,\ C,\ D$

> Lowering the **threshold** detects C but also flags D.

Visual Threshold Comparison

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Changing the threshold changes which cases are flagged

Same probabilities · different decision thresholds

Transaction	A	B	C	D	E	F
True class	Fraud	Legit.	Fraud	Legit.	Fraud	Legit.
Fraud score	0.91	0.64	0.48	0.22	0.18	0.07
t = 0.50	Flag	Flag	Pass	Pass	Pass	Pass
t = 0.20	Flag	Flag	Flag	Flag	Pass	Pass

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💡 At $t = 0.50$, only A and B are flagged. At $t = 0.20$, C and D join them: **one additional true positive and one additional false positive**.

Worked Threshold Metrics

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Compute precision and recall at two thresholds

Using the six labeled transactions A–F shown earlier:

Threshold	TP	FP	FN	TN	Precision	Recall
$t = 0.50$	1	1	2	2	$1/2 = 0.50$	$1/3 \approx 0.33$
$t = 0.20$	2	2	1	1	$2/4 = 0.50$	$2/3 \approx 0.67$

Lowering the **threshold** raises recall, **but precision stays the same in this example**.

> This is why we say precision *may* decrease as the threshold drops: it is **not guaranteed** to change in only one direction.

Use Only

Precision–Recall Trade-off

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How the decision threshold changes predictions

The **decision threshold** determines which predicted probabilities are classified as positive.

Lower threshold

- **More** examples predicted positive.
- More actual positives may be detected.
- **Recall cannot decrease** for fixed scores.
- False positives may increase.
- Precision may **increase or decrease**.

Higher threshold

- **Fewer** examples predicted positive.
- Predictions become more selective.
- **Recall cannot increase** for fixed scores.
- Some actual positives may be missed.
- Precision may **increase or decrease**.

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💡 **Key idea.** Lowering the threshold generally favors **recall**, while raising it makes positive predictions more selective–but **precision is not guaranteed to improve**. Choose the threshold using validation data and the costs of FP and FN.

ROC Curve

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True positive rate vs. false positive rate

The **ROC curve** shows how two rates change as the classification threshold varies:

$$\mathrm{TPR} = \frac{TP}{TP + FN}, \qquad \mathrm{FPR} = \frac{FP}{FP + TN}$$

- **True positive rate (TPR):** Proportion of actual positives correctly detected; also called **recall**.
- **False positive rate (FPR):** Proportion of actual negatives incorrectly classified as positive.

Interpreting ROC AUC

ROC AUC summarizes how well a model ranks positive examples above negative examples.

- **AUC near 1:** Strong ranking ability.
- **AUC near 0.5:** Similar to random ranking.
- **Higher AUC:** Generally better discrimination between classes.

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Class imbalance matters. ROC AUC may look favorable even when false alarms are operationally costly. For rare positive classes, also inspect **precision–recall performance**.

Precision–Recall Curve

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Often more informative for rare positives

The **precision–recall curve** shows the relationship between:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}$$

as the **decision threshold** changes.

Why is it useful?

- **Precision:** Of all flagged examples, how many are actually positive?
- **Recall:** Of all actual positives, how many are detected?

Precision–recall analysis is especially useful when:

- the positive class is **rare**;
- false positives and false negatives have different costs;
- detecting the positive class is the main objective.

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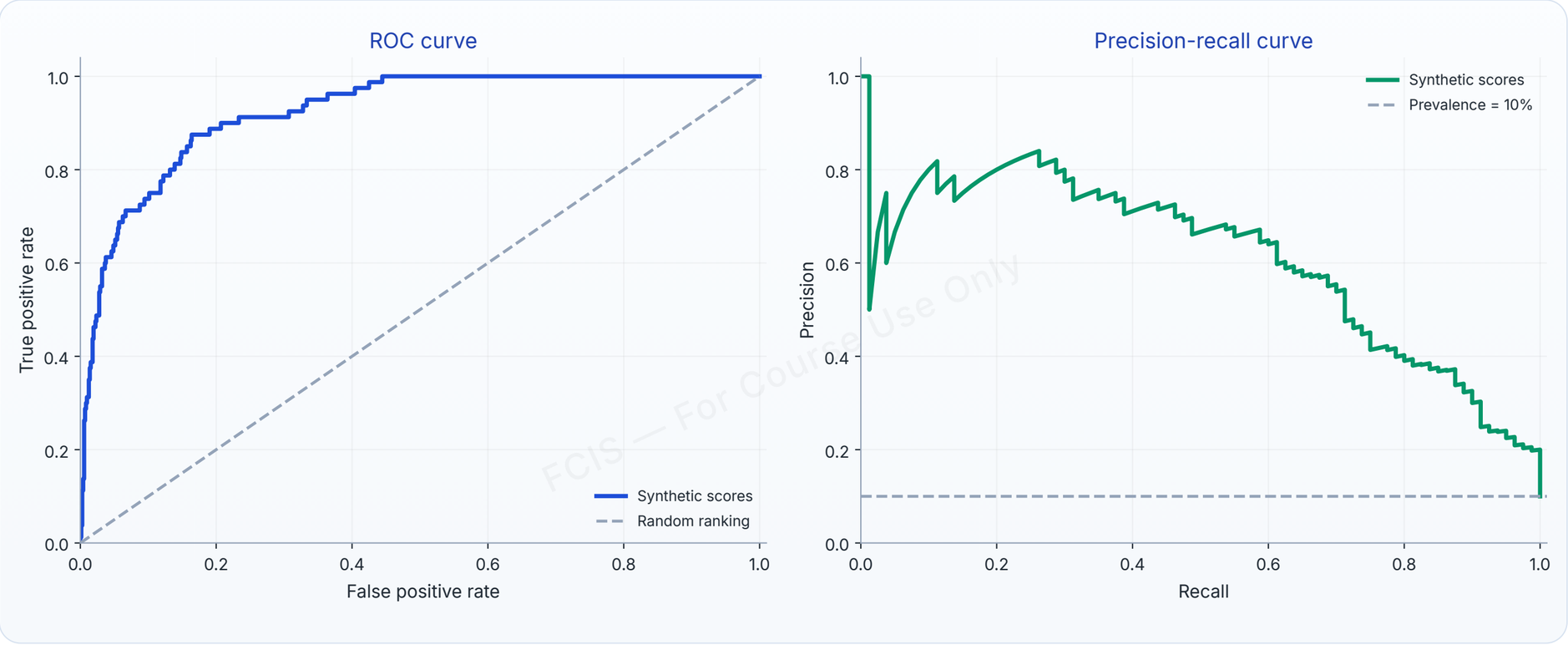
💡 **Key idea.** For tasks such as **fraud detection** or **rare disease diagnosis**, precision–recall analysis helps assess whether a model detects important cases without producing too many false alarms.

ROC and Precision–Recall Curves

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Same scores, different perspectives

A **synthetic dataset** contains 80 positives and 720 negatives (**10% prevalence**). Both curves are constructed from exactly the same scores as the decision threshold varies.



💡 **Key distinction.** ROC compares detection rate with the **false positive rate**. Precision–recall shows how many flagged examples are truly positive, making it especially useful when positives are rare.

Cost-Sensitive Evaluation

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Different errors have different consequences

In some applications, **false positives (FP)** and **false negatives (FN)** have very different costs.

Suppose:

- **False negative:** 500 SAR per missed positive case.
- **False positive:** 10 SAR per false alarm.

Total error cost

$$\boxed{Cost = 500(FN) + 10(FP)}$$

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A model with **lower accuracy** may still be preferable if it avoids expensive false negatives.

> 💡 **Key idea.** Model evaluation should reflect the **consequences of errors**, not just how many predictions are correct.

Worked Cost Example

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Two fraud models, same validation set

Both models are evaluated on **500 transactions**: 50 fraudulent and 450 legitimate.

Model	TP	FP	FN	TN	Accuracy
A	40	20	10	430	**94.0%**
B	47	60	3	390	**87.4%**

Assume **500 SAR per FN** and **10 SAR per FP**.

Compare total error costs

$$Cost_A = 10(500) + 20(10) = \mathbf{5{,}200}\ \text{SAR}$$

$$Cost_B = 3(500) + 60(10) = \mathbf{2{,}100}\ \text{SAR}$$

Decision: Model B has lower accuracy, but its estimated error cost is **3,100 SAR lower** because it misses fewer fraud cases.

💡 **Lesson.** The preferred model depends on the **costs of FP and FN**. Different cost assumptions could lead to a different choice.

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Part VI

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Validation and Model Selection

Building on the train/validation/test discipline from Lecture 1, we now **compare real candidates** and choose algorithms, features, and hyperparameters from validation evidence.

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What Is a Hyperparameter?

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Chosen before or outside training

Parameters (e.g., coefficients and weights) are **learned from training data**.

Hyperparameters control model structure or training and are chosen outside an individual training fit.

Model	Examples of hyperparameters
Ridge / Lasso	Regularization strength λ
Logistic regression	Regularization strength
KNN	Number of neighbors k
Decision tree	Maximum depth, minimum samples per leaf
Neural network	Learning rate, number of layers, hidden units
K-Means	Number of clusters K

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Selection matters. Tune supervised-model hyperparameters using **validation data or cross-validation**, never the final test set. A **classification threshold** is a separate decision choice: changing it alters predicted labels without retraining the model.

Train–Validation–Test Workflow

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Development first, final evaluation last

A reliable workflow separates **training**, **model selection**, and **final evaluation**.

1. Prepare & Train

- Split into train, validation, and test sets.
- Fit preprocessing on training data only.
- Train candidate models.

2. Validate & Select

- Compare models on validation data.
- Tune hyperparameters and, if needed, thresholds.
- Select the final model.

3. Finalize & Test

- Refit on train + validation if appropriate.
- Refit preprocessing with the model.
- Evaluate once on the untouched test set.

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💡 **Key idea.** Training data is used to learn the model, validation data guides model selection, and the **untouched test set** estimates performance on unseen data.

Cross-Validation

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More reliable comparison when data are limited

k-fold cross-validation evaluates a model across multiple training–validation splits.

1. Divide the **development data** into k folds.
2. Fit preprocessing and train the model on $k-1$ **folds**.
3. Evaluate on the **remaining validation fold**.
4. Repeat until every fold has served as validation, then **average the scores**.

Why use it?

Each example is used for **validation once** and for **training $k-1$ times**, making evaluation less dependent on one particular split.

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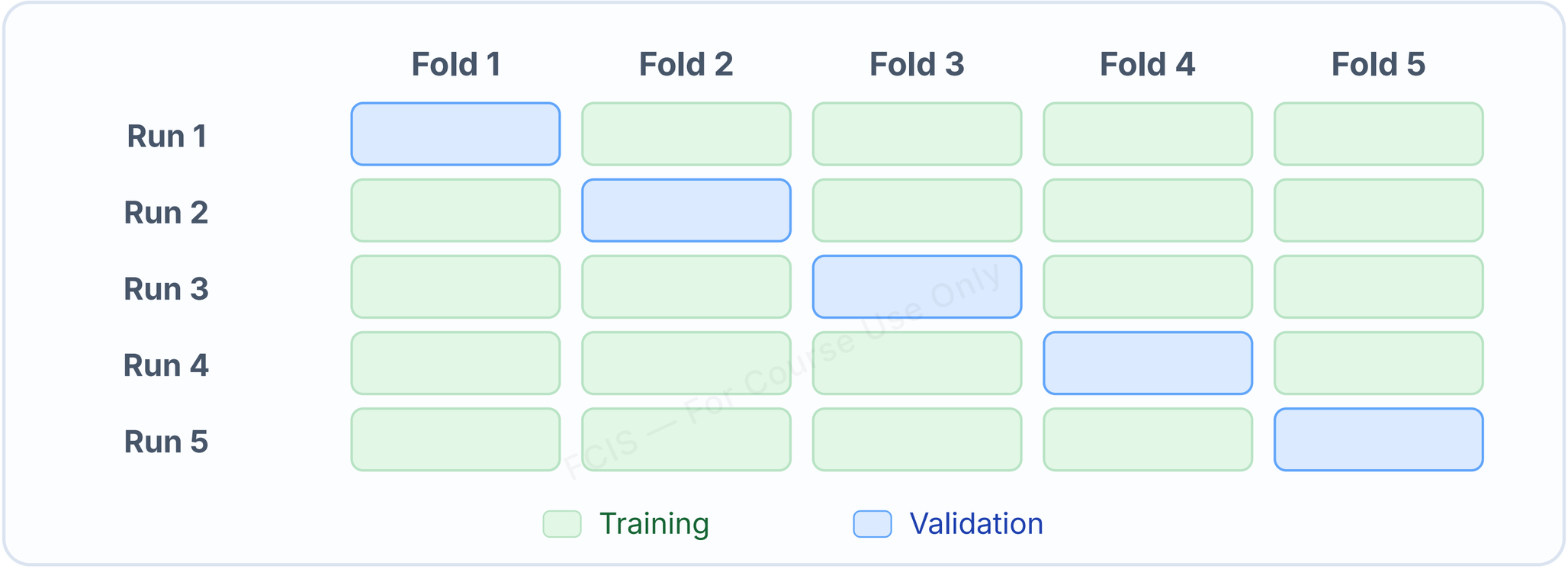
💡 **Key idea.** Use cross-validation to compare models and tune hyperparameters while keeping the **final test set untouched**. Choose stratified, group-aware, or time-aware folds when appropriate.

Cross-Validation at a Glance

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Five rounds, five different validation folds

In **5-fold cross-validation**, each run trains on four folds and validates on the fifth.



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Avoid leakage: Fit preprocessing separately on the training folds of each run. Never use the held-out fold to calculate scaling statistics or other learned transformations.

Cross-Validation Score

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Combine the results across folds

Suppose a KNN model with $k = 7$ **neighbors** produces these illustrative validation accuracies:

Validation fold	1	2	3	4	5
Accuracy	0.89	0.93	0.90	0.92	0.91

Average validation performance

$$\overline{\text{Accuracy}}_{\text{CV}} = \frac{0.89 + 0.93 + 0.90 + 0.92 + 0.91}{5} = \boxed{0.91}$$

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Interpretation. The mean cross-validation accuracy is **91%**. We report the average rather than selecting the best fold's score of 93%.

💡 Cross-validation produces **multiple validation scores**. Their mean summarizes performance, while their variation reveals sensitivity to the data split.

Model Selection Example

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Choosing k for KNN

Consider the following **illustrative 5-fold cross-validation results**.

k	Mean training accuracy	Mean CV accuracy
1	**1.00**	0.84
3	0.93	0.89
7	0.88	**0.91**
15	0.82	0.86

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Questions

1. Which k achieves the highest training accuracy?
2. Which k achieves the highest mean CV accuracy?
3. Which k should we select, and why?
4. What does the performance gap for $k = 1$ suggest?

Choosing k for KNN

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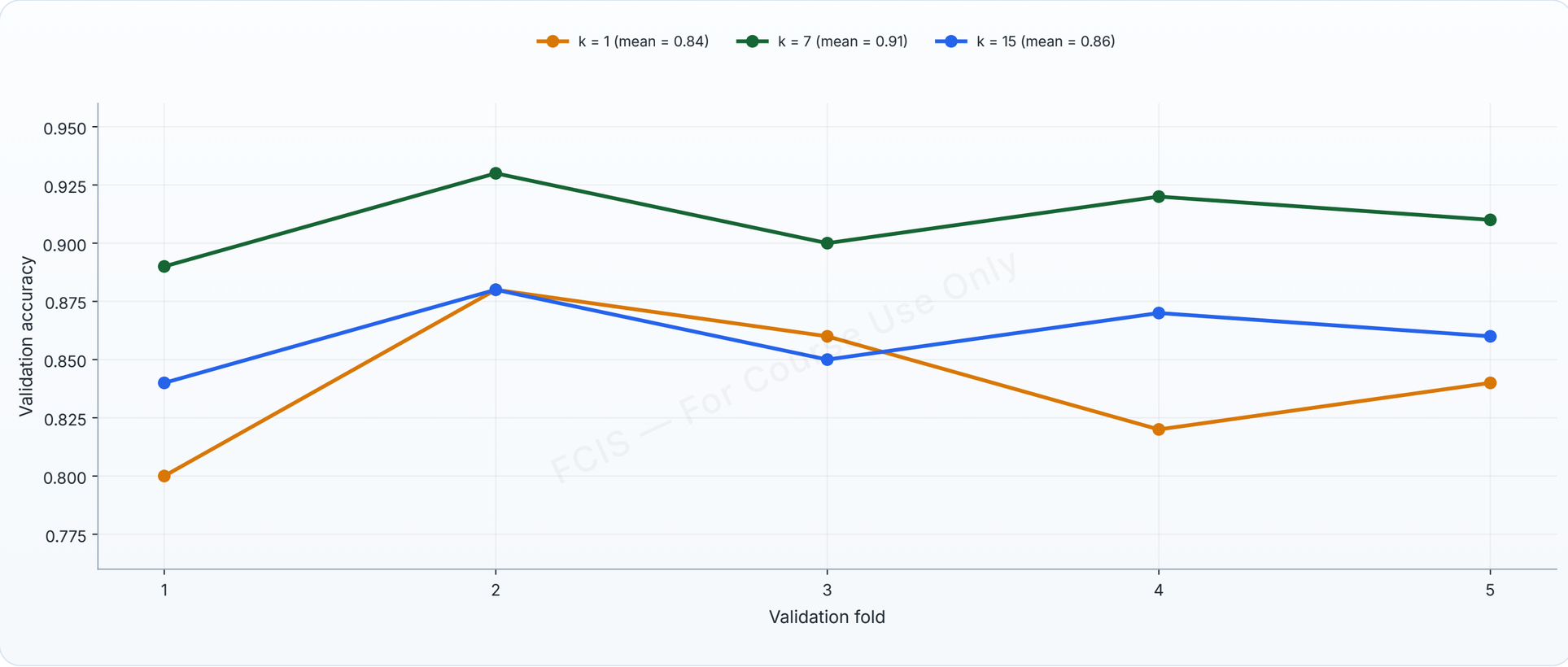
Solution: Select $k = 7$ because it achieves the highest mean CV accuracy (**91%**). For $k = 1$, perfect training accuracy but lower CV accuracy (**84%**) suggests **overfitting**.

Cross-Validation Variability

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A good mean can hide variation across folds

Illustrative five-fold validation accuracies for **three KNN settings** from the previous slide. These are example values, not measured experimental results.



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💡 **Key idea.** Compare both **mean validation performance** and **fold-to-fold variability**. Use cross-validation to select hyperparameters, then evaluate the finalized model on the untouched test set.

Nested Decision Discipline

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Avoid selecting from the test set

Incorrect workflow

Try many models
Compare candidates

→

Choose best test score
Test set influences selection

→

Report same test score
Optimistically biased result

Correct workflow

Try many models
Compare candidates

→

Choose using validation
Select the model

→

Evaluate on test set
Report final performance once

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Why?

Repeatedly using the test set to select models makes the reported performance **too optimistic**.

> **Takeaway.** Use validation for model selection; reserve the **test set** for final evaluation.

Part VII

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Worked Evaluation Case

Now we combine metrics, thresholds, and context in one classification example.

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Case: At-Risk Student Detection

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Problem setting

A university wants to identify students who may fail a course early enough to offer support.

Positive class:

$$y = 1 \quad \text{means at risk.}$$

Possible errors:

Error	Meaning	Consequence
False positive	student flagged but not actually at risk	unnecessary advising workload
False negative	at-risk student not flagged	missed opportunity for support

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💡 In this context, false negatives may be more serious than **false positives**.

Case: Two Models

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Validation results

Model	TP	FP	FN	TN
A	42	18	28	212
B	55	45	15	185

Compute

For each model:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Guiding question

Which model is more suitable if the main goal is to avoid missing at-risk students?

Case: Interpretation

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Metrics are connected to action

Metric	Model A	Model B
Precision	$\frac{42}{42+18} = 0.700$	$\frac{55}{55+45} = 0.550$
Recall	$\frac{42}{42+28} = 0.600$	$\frac{55}{55+15} \approx 0.786$
F_1 **score**	$\frac{2(42)}{2(42)+18+28} \approx 0.646$	$\frac{2(55)}{2(55)+45+15} \approx 0.647$
Students flagged	**60**	**100**

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Interpretation

- **Model A:** More precise alerts and fewer students flagged.
- **Model B:** Identifies more at-risk students (**55 vs. 42**) but flags **40 additional students**.
- Both models have nearly identical F_1 **scores**, despite different precision–recall trade-offs.

Decision. If avoiding missed at-risk students is the priority, **Model B** is preferable: it misses only **15** students instead of **28**, provided the additional advising workload is acceptable.

Case: Add Cost

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Make the decision explicit

Suppose:

- **false positive** cost = 1 advising unit;
- **false negative** cost = 5 missed-support units.

Model A:

$$Cost_A = 18(1) + 28(5) = 158$$

Model B:

$$Cost_B = 45(1) + 15(5) = 120$$

Decision

Model B has lower total cost under these assumptions.

> 💡 A model should be selected using the consequences of errors, not metric labels alone.

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Part VIII

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Common Evaluation Mistakes

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When good scores can be misleading

A high score does not always mean a good model.

Trustworthy evaluation requires the right data, metrics, and context.

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Mistake 1

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Evaluating on training data only

A student reports **99% training accuracy** and concludes that the model is excellent.

Why is this misleading?

Training accuracy measures performance on data the model has **already seen**. A high score may result from:

- memorizing training examples;
- fitting noise instead of general patterns;
- excessive model complexity;
- information leakage.

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A model can perform very well during training but poorly on **unseen data**.

> **Lesson.** Training accuracy alone does not demonstrate **generalization**. Evaluate using properly separated validation and test data.

Mistake 2

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Optimizing directly on the test set

A student repeatedly changes the:

- model type;
- selected features;
- hyperparameters;
- classification threshold;

until the **test score** becomes high.

Why is this a problem?

The test set is now influencing model selection. Repeated experimentation can lead to **overfitting the test set**, even without directly training on it.

> **Lesson.** Use **validation data or cross-validation** for model development. Reserve the **untouched test set** for final evaluation.

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Mistake 3

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Using the wrong metric

A rare-disease classifier achieves **98% accuracy**.

But suppose only **2% of patients** have the disease.

A model that predicts **“healthy” for everyone** also achieves:

- **Accuracy:** 98%
- **Recall for disease cases:** 0%
- **Detected disease cases:** None

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Better evaluation

Examine the **confusion matrix, precision, recall, false-negative costs**, and decision-threshold behavior.

> 💡 **Lesson.** A metric is useful only when it reflects the **goal and consequences** of the prediction task. High accuracy alone may hide critical failures.

Mistake 4

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Ignoring deployment context

Even a model with excellent validation results may be unsuitable in practice.

Questions to consider

- **Feature availability:** Will inputs be available at prediction time?
- **Decision costs:** Are predictions worth acting on?
- **Group performance:** Are errors distributed unequally?
- **Speed:** Can the model meet time constraints?
- **Explainability:** Are understandable decisions required?
- **Data shift:** Will future data resemble the evaluation data?

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💡 **Lesson.** A useful model must perform well **and** satisfy the requirements of its real-world application.

Practice & Consolidation

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Evaluation Decisions and Common Mistakes

How do we know whether a model is truly good?

Use the right metrics, validation strategy, and decision threshold—and consider the consequences of errors.

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Activity: Choose the Metric

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Match evaluation metrics to the task

For each application, choose suitable **evaluation metrics** and explain why they matter.

Task	Relevant metric(s)	Why?
Predict apartment rent		
Detect rare fraud		
Identify at-risk students		
Classify product category		
Forecast energy demand		
Medical screening		

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Match evaluation metrics to the task

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Detect rare fraud		
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Forecast energy demand		
Medical screening		

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Possible answers:

- **Rent / Energy:** MAE or RMSE to measure prediction error.
- **Fraud:** Precision, recall, and error costs.
- **At-risk students:** Recall, precision, and intervention workload.
- **Product category:** Accuracy, confusion matrix; macro-F_1 if classes are imbalanced.
- **Medical screening:** Sensitivity (recall), specificity, and error costs.

Remember: There may be more than one suitable metric.

Activity: Interpret a Confusion Matrix

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Calculate and explain the metrics

A binary classifier produces:

$$TP = 30, \quad FP = 15, \quad FN = 10, \quad TN = 145$$

Calculate

1. **Accuracy**
2. **Precision**
3. **Recall (sensitivity)**
4. **Specificity**
5. F_1 **score**

Interpret

> Is the model better at correctly rejecting **negative cases** or detecting **positive cases**? Compare specificity with recall.

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Calculate and explain the metrics

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Calculate

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3. **Recall (sensitivity)**
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5. F_1 **score**

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Interpret

Is the model better at correctly rejecting **negative cases** or detecting **positive cases**? Compare specificity with recall.

Solution:

- Accuracy = **87.5%**; Precision = **66.7%**; Recall = **75.0%**
- Specificity = **90.6%**; F_1 = **70.6%**

Interpretation: The model correctly rejects a greater proportion of negatives (**90.6%**) than the proportion of positives it detects (**75%**).

Activity: Threshold Reasoning

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Predict the effect before computing

A binary classifier changes its decision threshold from **0.5 to 0.3**, while keeping the predicted scores fixed.

Predict the effect of this change:

Quantity	Increase, decrease, or uncertain?
Number of predicted positives	
Recall	
False positives (FP)	
Precision	
False negatives (FN)	

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Number of predicted positives	
Recall	
False positives (FP)	
Precision	
False negatives (FN)	

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Solution:

- **Predicted positives:** Increase or remain unchanged.
- **Recall:** Increase or remain unchanged.
- **False positives:** Increase or remain unchanged.
- **Precision:** May increase or decrease (or remain unchanged).
- **False negatives:** Decrease or remain unchanged.

Why? A lower threshold flags more examples, but not all newly flagged examples are truly positive.

Activity: Select a Model

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Validation evidence and practical constraints

Three illustrative fraud-detection models are evaluated on the **same 1,000 transactions** (100 fraud, 900 legitimate).

Model	Accuracy	Precision	Recall	Flagged cases
A	**94.5%**	**0.88**	0.52	59
B	93.3%	0.62	**0.84**	135
C	94.0%	0.70	0.70	100

Questions

1. Which model would you choose for **fraud screening**, where missing fraud is costly?
2. Which model would suit a **limited manual review team**?
3. Which model has the most **balanced precision and recall**?
4. Why is the model with the highest accuracy not automatically best?

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Questions

1. Which model would you choose for **fraud screening**, where missing fraud is costly?
2. Which model would suit a **limited manual review team**?
3. Which model has the most **balanced precision and recall**?
4. Why is the model with the highest accuracy not automatically best?

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Possible decisions:

- **B:** Highest recall (**84%**); detects the most fraud, but flags more transactions.
- **A:** Highest precision (**88%**) and fewest flagged cases; suitable when review capacity is limited.
- **C:** Equal precision and recall (**70%**), offering a balance between them.

Lesson: Model selection depends on **missed-case costs, false alarms, and available resources**, not accuracy alone.

Mini-Quiz

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Check your understanding

1. Why should the final test set not be used for **hyperparameter tuning**?
2. What is the difference between precision and recall?
3. Why can accuracy be misleading for imbalanced data?
4. When is RMSE more informative than MAE?
5. What does $R^2 < 0$ indicate?
6. What usually happens to recall when the classification threshold is lowered?
7. Why can a model with lower accuracy still be better in a real application?
8. What is the purpose of cross-validation?

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Check your understanding

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Suggested answers: to keep final evaluation unbiased; precision measures correctness among predicted positives, recall measures coverage of actual positives; majority-class predictions can dominate accuracy; RMSE emphasizes large errors; worse than predicting the mean; recall usually increases; error costs may matter more; cross-validation gives more stable validation estimates.

Lecture 5 Summary

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Key ideas to retain

Evaluation Foundations

- **Training performance** alone does not demonstrate generalization.
- **Validation data** guides model selection and hyperparameter tuning.
- The **untouched test set** provides the final performance estimate.
- **Baselines** show whether a model offers meaningful improvement.
- **Data leakage** can produce misleadingly optimistic results.
- **Cross-validation** reduces dependence on a single validation split.

Metrics and Decisions

- **Regression metrics** measure different aspects of prediction error.
- **Confusion matrices** distinguish TP, FP, FN, and TN.
- **Precision, recall, specificity, and F_1** emphasize different outcomes.
- **Class imbalance** can make accuracy misleading.
- **Decision thresholds** affect recall, precision, and error trade-offs.
- **Error costs and application context** should guide model selection.

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End-of-Lecture Check

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Five questions to test your understanding

1. What are the different roles of **training, validation, and test data**?
2. Why should we compare a model against a **baseline**, and how can data leakage invalidate the comparison?
3. Which metric would you prioritize when **false negatives are very costly**, and why?
4. What happens to **recall and false positives** when the classification threshold decreases? Is precision guaranteed to decrease?
5. Why might a model with **lower accuracy** be preferable when real-world error costs are considered?

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💡 **Preparation for Lecture 6.** Review **entropy, information gain, feature-based decision rules, Euclidean distance, and feature scaling**. Next, we explore **decision trees and KNN**—two different approaches to making predictions.

Suggested Reading

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Primary and supporting references

Primary reference

G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review sections on assessment of model accuracy, validation, cross-validation, classification metrics, and model selection.

Supporting references

A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review classification evaluation, confusion matrices, precision/recall, ROC curves, cross-validation, and model tuning.

K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for deeper views of predictive uncertainty, losses, and decision-theoretic evaluation.

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💡 When reading, focus on why a metric is appropriate for a particular decision, not only how to calculate it.